# Quaternionic Logic: A G-Lattice Unifying Boolean and Fuzzy Frameworks

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**Abstract**

A layered ontology of 72 semantic primitives across six algebraic layers—Boolean, Fuzzy, Ordinal, Modal, Trivalent, and Probabilistic—was previously shown to discriminate real knowledge domains from pseudoscience controls. However, the six algebras remained independent systems with no formal unification.

We construct a single quaternionic operator $\mathcal{O} = \{0, 1, +, i, j, k\}$ from which all six algebras admit dimensional projections—four (Boolean, Gödel fuzzy, Gödel–Kripke modal, and Kleene trivalent) as exact recoveries, two (ordinal and probabilistic) as interpretive correspondences. The operator assigns to each concept a four-dimensional coordinate $(r, i, j, k) \in [0, 1]^4$, where $r$ encodes crystallised truth, $i$ encodes potentiality, $j$ encodes verifiability, and $k$ encodes recursive selection. Five axioms (K1–K5) characterise $k$, and its *irreducibility* is proven: no function of $(r, i, j)$ can reproduce $k$.

Logical connectives on $V = [0, 1] \times [-1, 1]^3$ form a bounded residuated lattice with involutive negation. An SU(2) group acts on truth values by conjugation, defining context transformations that preserve truth while rotating epistemic character. The central result is a *commutation theorem*: lattice connectives commute with the group action if and only if they depend exclusively on the real component. Boolean logic emerges as the *context-free* (group-invariant) special case.

The combined structure is a *G-lattice*: a residuated lattice equipped with a non-abelian group action. Additional results include: a cancellative–generative classification of nine algebras of opposition, four diagnosable pathologies as axis collapses, and a *liquid zone* (all four axes positive) shown to be open, convex, and path-connected. Completeness reduces to $G^{\sim}$ (Gödel logic with involutive negation). All results are verified computationally at machine precision ($\sim 10^{-16}$) over 50,000 test cases.

## 1 Introduction

Six algebraic layers were introduced in Ornelas Brand [2026c] to organise the 72 semantic primitives of the prime-algebraic neurosymbolic bridge [Ornelas Brand, 2026a] into a dependency DAG: Boolean ($\{0, 1\}$), Fuzzy ($\{0, +\}$), Ordinal ($\{<, =, >\}$), Modal ($\{\Diamond, \Box\}$), Trivalent ($\{-, 0, +\}$), and Probabilistic ($\{0, ?\}$). Each layer has its own formal logic, its own operators, and its own domain of application. Yet they remained independent systems.

This paper asks two questions. First: *is there a single operator from which all six are recoverable as projections?* Second: *what formal logic does that operator support?*

We answer both affirmatively. The operator $\mathcal{O} = \{0, 1, +, i, j, k\}$ assigns to each semantic state a point in a four-dimensional unit hypercube, where each axis corresponds to an irreducible ontological category: crystallised truth ($r$), potentiality ($i$), verifiability ($j$), and recursive selection ($k$). The six algebras are shadows cast by this four-dimensional object onto lower-dimensional subspaces. The operator supports a residuated lattice equipped with an SU(2)

group action—a structure we call a *G-lattice*—in which Boolean logic emerges as the unique context-free special case.

**Contributions.**

1. Construction of a quaternionic operator from which all six algebraic layers are recoverable—four exactly (Boolean, Gödel fuzzy, Gödel–Kripke modal, Kleene trivalent) and two interpretively (Sections 3–7).
2. Five axioms characterising the fourth axis $k$, with a proof of its irreducibility (Sections 4, 11).
3. A bounded residuated lattice on $[0,1] \times [-1,1]^3$ with exact recovery of four formal logics (Section 6).
4. A commutation theorem characterising Boolean logic as the group-invariant sublattice (Section 8).
5. Soundness and completeness, the latter reducing to $G^{\sim}$ (Gödel logic with involutive negation) via a real-part independence lemma (Sections 9, 10).
6. A cancellative–generative classification of nine algebras of opposition, with formal $k$-embeddings for all five generative algebras (Section 12).
7. Four diagnosable pathologies as axis collapses, with a liquid zone shown to be open, convex, and path-connected (Section 13).
8. A layer-based embedding of the prime algebra into the G-lattice domain, with a bilateral synthesis theorem characterising strict increase (Section 18).
9. A bilattice embedding of Belnap's **FOUR** into the G-lattice, preserving both truth and information orders as lattice homomorphisms (Section 7).
10. Computational verification of all results at machine precision (Section 19).

# 2 Preliminaries

## 2.1 Quaternions

The quaternion algebra $\mathbb{H}$ is the four-dimensional real algebra spanned by $\{1, i, j, k\}$ with multiplication rules $i^2 = j^2 = k^2 = ijk = -1$. Every quaternion $q = r + ai + bj + ck$ has real part $\mathrm{Re}(q) = r$ and imaginary part $\mathrm{Im}(q) = (a, b, c)$. The conjugate is $\bar{q} = r - ai - bj - ck$, and for unit quaternions ($|q| = 1$), $q^{-1} = \bar{q}$.

The group SU(2) of unit quaternions acts on $\mathbb{H}$ by conjugation: $q \cdot a = q\,a\,\bar{q}$. This action preserves the real part, the imaginary norm, and the total norm:

$$\mathrm{Re}(q\,a\,\bar{q}) = \mathrm{Re}(a), \tag{1}$$

$$\|\,\mathrm{Im}(q\,a\,\bar{q})\| = \|\,\mathrm{Im}(a)\|, \tag{2}$$

$$\|q\,a\,\bar{q}\| = \|a\|. \tag{3}$$

## 2.2 Residuated lattices

A *bounded residuated lattice* is a structure $(L, \wedge, \vee, \otimes, \rightarrow, 0, 1)$ where $(L, \wedge, \vee, 0, 1)$ is a bounded lattice, $\otimes$ is a monoidal operation (the t-norm), and $\rightarrow$ is its residuum satisfying $a \otimes b \leq c \iff a \leq b \rightarrow c$ [Galatos et al., 2007].

# 3 Construction of the Operator

The construction proceeds in four steps, each adding one axis that no previous combination could capture.

### 3.1 Step 1: Recovering Boolean and Fuzzy — the $r$ axis

The simplest attempt to subsume $\{0, 1\}$ and $\{0, +\}$ is to take their intersection: $\{0, +\}$, where $+$ abstracts over both 1 (discrete) and the open interval $(0, 1)$ (continuous). But this loses information: from $\{0, +\}$ one cannot recover which system was in use.

A true unifier must preserve both as special cases:

$$\mathcal{O}_1 = \{0,\ 1,\ +\} \tag{4}$$

where 0 is absence, 1 is resolved (categorical) presence, and $+$ is partial (graded) presence. Recovery is immediate:

$$\text{Boolean} = \mathcal{O}_1\big|_{+=\varnothing} = \{0, 1\} \tag{5}$$

$$\text{Fuzzy} = \mathcal{O}_1 \text{ with } + \in (0, 1),\ 1 = \sup \tag{6}$$

This defines the first axis: $r \in [0, 1]$, the degree of *crystallised presence*. A concept with $r = 1$ is fully resolved; with $r = 0.3$, partially so.

### 3.2 Step 2: Incorporating modality — the $i$ axis

The modal algebra $\{\Diamond, \Box\}$ distinguishes possibility from necessity. Necessity ($\Box p$) maps to $r = 1$: the concept is fully crystallised with no residual potential. But possibility ($\Diamond p$) introduces a genuinely new dimension: something that is *neither present nor absent* but *latent*—potential that has not yet actualised.

No value of $r$ captures this. Adding a second axis:

$$\mathcal{O}_2 = \{0,\ 1,\ +,\ i\} \tag{7}$$

with the modal correspondence:

$$\Box p \longleftrightarrow (r = 1,\ i = 0) \tag{8}$$

$$\Diamond p \longleftrightarrow (r \geq 0,\ i > 0) \tag{9}$$

Both $r$ and $i$ admit continuous degrees, yielding a *semantic complex plane* where each state is a pair $(r, i) \in [0, 1]^2$.

### 3.3 Step 3: The polarity axis — $j$

The trivalent algebra $\{-, 0, +\}$ introduces *polarity*: not just presence or absence, but *direction*. This is irreducible to $r$ (which has no sign) and to $i$ (which has no directionality).

$$\mathcal{O}_3 = \{0,\ 1,\ +,\ i,\ j\} \tag{10}$$

The $j$ axis captures the degree of *verifiability and directional comparability*. It subsumes the ordinal algebra $\{<, =, >\}$, which are relational operators that become possible only when a measurement axis exists.

### 3.4 Step 4: Recursive selection — the $k$ axis

Three axes generate a static space: concepts have crystallisation, potentiality, and directionality, but nothing *moves*. The system cannot prune, cannot advance, cannot select.

The probabilistic algebra $\{0, ?\}$ of Layer 6 (Observer) hints at this missing dimension. The "?" is not mere uncertainty—it is the state of a system that *looks at itself* and does not yet know what it will find. This self-referential act—observation as recursive selection—is irreducible to the other three axes.

The complete operator is:

$$\boxed{\mathcal{O} = \{0,\ 1,\ +,\ i,\ j,\ k\}} \tag{11}$$

where every semantic state lives in:

$$s = (r,\ i,\ j,\ k) \in [0,1]^4 \tag{12}$$

The $k$ axis measures the degree of *recursive selection*: how much a system prunes itself, rotates its perspective, and advances irreversibly toward higher complexity. The four axes are irreducible: no subset of three can generate the fourth. This mirrors the algebraic structure of the quaternions $\mathbb{H}$, where $\{1, i, j, k\}$ are the four basis elements and no three suffice to span the algebra. A formal proof is given in Section 11.

## 4 Axioms for Recursive Selection

We propose five axioms that characterise $k$.

**Axiom K1 — Range**

**Axiom 1** (Range)**.** $k \in [0,1]$ *for every semantic state. The value* $k = 0$ *represents the complete absence of selection;* $k = 1$ *represents maximal selective activity.*

This is a structural axiom ensuring $k$ lives in the same bounded interval as the other axes, consistent with the hypercube $H = [0,1]^4$.

**Axiom K2 — Stagnation Existence**

**Axiom 2** (Stagnation existence)**.** *The* stagnation hyperplane

$$\mathcal{S} = \{(r, i, j, 0) \in H : r > 0,\ i > 0,\ j > 0\}$$

*is inhabited. That is, there exist semantic states with all three of crystallisation, potentiality, and verifiability active, yet with zero recursive selection.*

**Empirical grounding.** Consider Kodak in 1975: the company had crystallised knowledge of photography ($r > 0$), imagined digital photography and built prototypes ($i > 0$), and verified feasibility with working systems ($j > 0$)—yet its membrane sealed and no selection occurred ($k = 0$). The company stagnated and eventually perished from conceptual heat death.

**Axiom K3 — Generation**

**Axiom 3** (Generation)**.** *The interaction of potentiality and verifiability* enables *but does not* determine *selection:*

$$i > 0 \ \wedge\ j > 0 \quad \Longrightarrow \quad k \text{ can be activated,}$$

*but there exist states with* $i > 0$, $j > 0$, *and* $k = 0$.

This axiom formalises the quaternionic relation $ij = k$ as an *enabling* relation rather than a deterministic one. In standard quaternion algebra, $ij = k$ is an identity. Here, it states that having both potentiality and verifiability is a *necessary precondition* for selection, but not sufficient—a system may have both and still fail to select (Axiom 2).

**Axiom K4 — Self-Inversion**

**Axiom 4** (Self-inversion). *Recursive selection applied to itself produces inversion. In the quaternion algebra $\mathbb{H}$, the basis element $k$ satisfies $k^2 = -1$: left-multiplication by $k$ twice reverses orientation.*

**Scope and domain.** K4 is a statement about the algebraic structure of $\mathbb{H}$, not about the semantic hypercube $H = [0,1]^4$: the map $-\mathrm{Id} : a \mapsto -a$ sends $H$ outside itself (e.g., $-(0.5, 0.3, 0.2, 0.1) \notin [0,1]^4$). K4 therefore lives in $\mathbb{H}$ and constrains the *algebraic substrate* that the semantic axes mirror, not the bounded domain of truth values.

**Left-multiplication vs. conjugation.** The G-lattice (Section 8) uses conjugation $q \cdot a = qa\bar{q}$, not left-multiplication $a \mapsto ka$. These are distinct actions with different iteration behaviour:

- *Left-multiplication:* $(a \mapsto ka) \circ (a \mapsto ka) = (a \mapsto k^2 a) = (a \mapsto -a)$. Self-inversion.
- *Conjugation:* $(a \mapsto ka\bar{k}) \circ (a \mapsto ka\bar{k}) = (a \mapsto k^2 a\bar{k}^2) = (a \mapsto (-1)a(-1)) = (a \mapsto a)$. Identity.

K4 captures the left-multiplication structure; the G-lattice action captures the conjugation structure. This duality is not an artefact of our construction: it is the standard distinction in Clifford algebra $\mathrm{Cl}(0,2) \cong \mathbb{H}$ between the left regular representation (spinor transformation) and the twisted adjoint representation (vector rotation) [Lounesto, 2001, Porteous, 1995]. In the Clifford framework, $\mathbb{H}$ as an $(\mathbb{H}, \mathbb{H})$-bimodule encodes both actions: left multiplication is the left module structure, and conjugation arises from the bimodule structure via the canonical anti-involution. The two actions coexist as structural features of the same algebra; neither is derivable from the other.

**Semantic reading.** A system that selects *only which selections to make*—recursion without base case—enters infinite regress and inverts. This is the algebraic signature of Hofstadter's strange loop [Hofstadter, 1979]: self-reference that, taken to its logical conclusion, flips the system. Remark 36 below discusses how this quaternionic form relates to the weaker involutive form found in categorical realisations of $k$.

**Axiom K5 — Activation Threshold**

**Axiom 5** (Activation threshold). *There exists a measurable function $\tau : (0,1]^3 \to [0,1]$ such that the transition from $k = 0$ to $k > 0$ occurs when the system's state $(r, i, j)$ crosses a critical surface:*

$$k \text{ activates when } \Psi(r,i,j) \geq \tau(r,i,j),$$

*where $\Psi$ is a system-dependent potential function.*

K5 is intentionally the weakest axiom: it postulates the *existence* of a threshold without specifying its form. This generality is deliberate—different substrates have different activation dynamics, and any stronger constraint (e.g., requiring $\tau$ to be continuous, monotone, or given by a specific formula) would restrict the framework to a particular substrate class. The axiom is not vacuous: it constrains $k$-activation to depend on a *deterministic* surface in $(r,i,j)$-space, excluding stochastic or history-dependent activation. It is non-trivially satisfiable: any continuous $\Psi$ with $\Psi(r,i,j) = rij$ and any constant $\tau_0 \in (0,1)$ produce a concrete threshold surface $rij = \tau_0$.

**Remark 1** (Mutual consistency). *The five axioms are mutually consistent: K1 bounds $k$; K2 guarantees stagnation exists; K3 says $i$ and $j$ enable but don't force $k$; K4 characterises pure $k$'s self-action; K5 says activation has a threshold. No pair contradicts another, and K2's second clause is implied by K3's second clause (making K2 the stronger existential statement while K3 is the structural one).*

## 5 The Four Kingdoms

Each axis corresponds to an ontological kingdom—an irreducible mode in which information can exist.

**Axis $r$ — The Conceptual**

What crystallises, fixes, defines. A concept with $r = 1$ is categorical: triangle, prime number, conservation law. It has found its form and holds it. The degree of resolved presence. *Ice.*

**Axis $i$ — The Imaginary**

What flows, invents, projects. A concept with $i = 1$ is pure potentiality: the "what if...?" that has not yet crystallised. It is real, vivid, potent—but without form. *Plasma.*

**Axis $j$ — The Exact**

What measures, compares, orders. A concept with $j = 1$ is fully verifiable: the demonstration, the experiment, the measurement that closes with *Q.E.D.* and admits no debate. The scale without temperature. *The thermometer.*

**Axis $k$ — The Recursive**

What selects, prunes, rotates, and advances. A concept with $k = 1$ is maximally recursive: it observes itself, discards what does not work, and irreversibly increases complexity. It is the engine of evolution, the phase transition, the moment where the system reorganises. *The spin.*

The four kingdoms are to information what the four fundamental forces are to physics: irreducible, complementary, and jointly exhaustive. No kingdom can be derived from the other three.

The state-of-matter metaphors—*ice* for $r$, *plasma* for $i$, *thermometer* for $j$, *spin* for $k$—are offered as *mnemonic*, not definitional: they are meant to make the ontological content easier to remember, not to ground it. The definitional content is carried by the axioms of Section 4 and the construction of Section 3, not by the metaphors.

## 6 The Lattice of Connectives

**Definition 2** (Truth values). *A* quaternionic truth value *is an element* $a = (r, i, j, k) \in V = [0,1] \times [-1,1]^3$. *The component* $r \in [0,1]$ *is the* degree of truth. *The triple* $(i, j, k) \in [-1,1]^3$ *is the* epistemic character.

**Definition 3** (Connectives). *Define component-wise operations on* $V = [0,1] \times [-1,1]^3$*:*

$$\mathrm{AND}(a,b) = (\min(a_r, b_r),\ \min(a_i, b_i),\ \min(a_j, b_j),\ \min(a_k, b_k)), \tag{13}$$

$$\mathrm{OR}(a,b) = (\max(a_r, b_r),\ \max(a_i, b_i),\ \max(a_j, b_j),\ \max(a_k, b_k)), \tag{14}$$

$$\mathrm{NOT}(a) = (1 - a_r,\ -a_i,\ -a_j,\ -a_k), \tag{15}$$

$$(a \to b)_m = \begin{cases} 1 & \text{if } a_m \le b_m, \\ b_m & \text{otherwise,} \end{cases} \quad \text{for each component } m \in \{r, i, j, k\}. \tag{16}$$

**Proposition 4** (Lattice structure). $(V, \mathrm{AND}, \mathrm{OR}, \to, \mathbf{0}, \mathbf{1})$ *is a bounded residuated lattice with bounds* $\mathbf{0} = (0, -1, -1, -1)$ *and* $\mathbf{1} = (1, 1, 1, 1)$ *under the component-wise order.*

*Proof.* Component-wise min and max on the bounded intervals $[0,1] \times [-1,1]^3$ form a bounded distributive lattice with $\mathbf{0} \leq a \leq \mathbf{1}$ for all $a \in V$.

*Residuation.* We verify $\mathrm{AND}(a,b) \leq c$ iff $a \leq (b \to c)$ for all $a,b,c \in V$, component by component. Fix any component $m$. ($\Rightarrow$): If $b_m \leq c_m$, then $(b \to c)_m = 1 \geq a_m$. If $b_m > c_m$, then $(b \to c)_m = c_m$ and we need $a_m \leq c_m$. From $\min(a_m, b_m) \leq c_m$ and $b_m > c_m$ it follows that $a_m \leq b_m$, hence $\min(a_m, b_m) = a_m \leq c_m$. ($\Leftarrow$): If $b_m \leq c_m$, then $\min(a_m, b_m) \leq b_m \leq c_m$. If $b_m > c_m$, then $a_m \leq (b \to c)_m = c_m$, so $\min(a_m, b_m) \leq a_m \leq c_m$. □

**Remark 5** (Involutive negation)**.** *The negation* (15) *is* not *the lattice negation* $a \to \mathbf{0}$ *(which is Gödel negation:* 1 *if* $a = \mathbf{0}$*,* $\mathbf{0}$ *otherwise). It is an additional* involutive *negation satisfying* $\mathrm{NOT}(\mathrm{NOT}(a)) = a$ *and the De Morgan laws:* $\mathrm{NOT}(\mathrm{AND}(a,b)) = \mathrm{OR}(\mathrm{NOT}(a), \mathrm{NOT}(b))$*, since* $1 - \min(x,y) = \max(1-x, 1-y)$ *and* $-\min(x,y) = \max(-x,-y)$*. The expanded structure is a residuated lattice with involutive negation in the sense of Esteva et al. [2000].*

**Remark 6** (Product structure of the lattice)**.** *The lattice* $(V, \mathrm{AND}, \mathrm{OR}, \to, \mathrm{NOT})$ *is isomorphic to the product of four independent chains:* $([0,1], \min, \max, \to_G, 1-x) \times ([-1,1], \min, \max, \to'_G, -x)^3$*. In particular, the connectives do not couple the four components. The quaternionic structure of the framework manifests not in the lattice connectives but in two other places: (i) the* $\mathrm{SU}(2)$ *group action (Section 8), which couples the imaginary components via non-abelian rotation, and (ii) the quaternionic product on* $S^3$ *(Section 14), which couples all four components via Hamilton multiplication. A non-product residuated lattice whose monoidal operation uses quaternion multiplication directly is not available: quaternion multiplication is not commutative, not idempotent, and does not preserve* $[0,1]^4$*. The construction of a non-commutative residuated lattice on* $V$ *using a quaternion-derived t-norm faces deep obstructions: six families of lattice orders are incompatible with the Hamilton product, and at most two of {associativity, non-commutativity, compatible lattice order} can coexist on* $V$*—a theorem, not a conjecture, proved via angular accumulation of iterated products [Ornelas Brand, 2026d].*

# 7 Recovery of Formal Systems

**Theorem 7** (Exact recovery)**.** *Under dimensional restrictions, the structure* $(V, \mathrm{AND}, \mathrm{OR}, \mathrm{NOT}, \to)$ *reduces exactly to:*

1. ***Boolean****:* $i = j = k = 0$*,* $r \in \{0,1\} \Rightarrow$ *classical propositional logic.*
2. ***Gödel fuzzy****:* $i = j = k = 0$*,* $r \in [0,1] \Rightarrow$ *Gödel fuzzy logic (min t-norm).*

*Proof.* Under restriction (1), $V$ collapses to $\{0,1\}$: min is Boolean AND, max is Boolean OR, $1-r$ is classical negation, and the residuum gives material implication. Under (2), $V$ collapses to $[0,1]$ with the standard Gödel connectives [Hájek, 1998]. Both are isomorphisms, not merely embeddings. □

## 7.1 Unified table of the six layers

The six algebraic layers introduced in Ornelas Brand [2026c] are projections of the operator $\mathcal{O}$:

Table 1: Six algebraic layers as projections of $\mathcal{O}$.

| Layer | Name | Original algebra | Unified operator | Active axes |
|---|---|---|---|---|
| 1 | Point | $\{0, 1\}$ | $\{0,\ 1\}$ | $r$ discrete |
| 2 | Line | $\{0, +\}$ | $\{0,\ +\}$ | $r$ continuous |
| 3 | Time | $\{<, =, >\}$ | $\{<,\ =,\ >\}$ | $j$ pure |
| 4 | Plane | $\{\Diamond, \Box\}$ | $\{i,\ 1\}$ | $r \times i$ |
| 5 | Volume | $\{-, 0, +\}$ | $\{-j,\ 0,\ +j\}$ | $j^{\pm}$ (polarity) |
| 6 | Observer | $\{0, ?\}$ | $\{0,\ rijk\}$ | $r \times i \times j \times k$ |

The layers progressively activate axes: the first two operate on $r$ alone (presence), the middle layers activate one or two additional axes ($i$, $j$), and Layer 6 (Observer) is the only one inhabiting all four dimensions simultaneously. Layers 1–5 set $k = 0$ (they predate selection).

**Theorem 8** (Modal and trivalent recovery)**.** *Under dimensional restrictions, two further logics are exactly recoverable:*

3. ***Modal (Gödel–Kripke)**: Restrict to $j = k = 0$, $r, i \in [0, 1]$, and equip the Gödel restriction with Kripke frames valued in $[0, 1]$, reading $\Box\varphi$ as the infimum of $\varphi$ over accessible worlds and $\Diamond\varphi$ as the supremum. Then the $\{0, 1\}$-subalgebra recovers classical modal logic exactly. Completeness of Gödel modal logics (including S4 and S5 at the Boolean restriction) is established by Caicedo & Rodriguez [2010]; the general construction for residuated lattices is due to Bou et al. [2011] and Fitting [1991].*
4. ***Trivalent (Kleene)**: Restrict to $i = k = 0$, $j \in \{-1, 0, +1\}$. The connectives* min, max, *and $-j$ on the three-element chain $\{-1, 0, +1\}$ are exactly the strong Kleene truth tables (with the identification $-1 =$ false, $0 =$ unknown, $+1 =$ true). The restriction is closed under all three operations and constitutes an isomorphism, not an approximation. More generally, Łukasiewicz $n$-valued logic is exactly the $n$-element subalgebra of the standard MV-algebra on $[0, 1]$ [Cignoli et al., 2000].*

*Proof. (3) Modal.* The Gödel chain $([0, 1], \min, \max, \to_G)$ is a Heyting algebra and hence admits Kripke semantics. By Caicedo & Rodriguez [2010], Theorem 3.8, the standard Gödel modal logic over $[0, 1]$ is complete with respect to Kripke frames with $[0, 1]$-valued accessibility. When the valuation is restricted to $\{0, 1\}$, the accessibility relation becomes crisp, the Gödel residuum becomes material implication, and the resulting logic is classical modal logic (K, S4, or S5 depending on frame conditions).

*(4) Trivalent.* On $\{-1, 0, +1\}$ with the natural order $-1 < 0 < +1$: min is Kleene conjunction, max is Kleene disjunction, and $x \mapsto -x$ is Kleene negation. These are exactly the strong Kleene truth tables. The subset is closed under all three operations (trivially: min, max, and negation of elements of $\{-1, 0, +1\}$ remain in $\{-1, 0, +1\}$). This gives an isomorphism between the restricted sublattice and $K_3$. $\Box$

**Remark 9** (Remaining interpretive correspondences)**.** *Two layers remain as interpretive correspondences rather than exact isomorphisms:*

5. ***Ordinal**: $j = k = 0$, $i$ encoding rank.*
6. ***Probabilistic**: $j = k = 0$, $i$ encoding variance.*

*These are suggestive readings of the restricted subspaces. Ordinal and probabilistic logics have richer structure (well-orders, $\sigma$-algebras) than the quaternionic restriction provides.*

### 7.2 Belnap–Dunn bilattice embedding

Belnap's four-valued logic [Belnap, 1977] uses the bilattice $\mathbf{FOUR} = \{\bot, \mathbf{f}, \mathbf{t}, \top\}$ with two independent lattice orders: a *truth* order ($\mathbf{f} \leq_t \bot \leq_t \mathbf{t}$, $\mathbf{f} \leq_t \top \leq_t \mathbf{t}$, with $\bot$ and $\top$ incomparable) and an *information* order ($\bot \leq_i \mathbf{f} \leq_i \top$, $\bot \leq_i \mathbf{t} \leq_i \top$, with $\mathbf{f}$ and $\mathbf{t}$ incomparable).

**Theorem 10** (Bilattice embedding)**.** $\mathbf{FOUR}$ *embeds into* $V = [0,1] \times [-1,1]^3$ *as a simultaneous lattice homomorphism for both orders. Specifically, there exists* $\varphi : \mathbf{FOUR} \to V$ *such that:*

1. $\varphi$ *preserves the truth order as the component-wise order on* $V$*:* $a \leq_t b$ *iff* $\varphi(a) \leq_{\mathrm{cw}} \varphi(b)$*.*
2. $\varphi$ *preserves the information order as the component-wise order on absolute imaginary magnitudes:* $a \leq_i b$ *iff* $|\mathrm{Im}(\varphi(a))| \leq_{\mathrm{cw}} |\mathrm{Im}(\varphi(b))|$*.*
3. *Both truth and information lattice operations (meet and join) are preserved.*

*Proof.* Choose parameters $r_0 < r_1$ in $[0,1]$, $s < 0 < t$ in $[-1,1]$ with $|s| > t$, and $0 < d$ in $[0,1]$. Define:

$$\varphi(\bot) = (r_0,\ t,\ 0,\ 0), \qquad \varphi(\mathbf{f}) = (r_0,\ s,\ 0,\ 0),$$
$$\varphi(\mathbf{t}) = (r_1,\ t,\ d,\ 0), \qquad \varphi(\top) = (r_1,\ s,\ d,\ 0).$$

The construction assigns each coordinate to either case (A) ($\bot_x = \mathbf{f}_x$, $\top_x = \mathbf{t}_x$) or case (B) ($\bot_x = \mathbf{t}_x$, $\top_x = \mathbf{f}_x$): $r$ takes case (A), $i$ takes case (B), and $j$ takes case (A).

*(1) Truth order.* Since $s < t$ and $r_0 < r_1$, the component-wise relations are: $\varphi(\mathbf{f}) \leq \varphi(\bot)$ (only $i$ changes: $s \leq t$), $\varphi(\bot) \leq \varphi(\mathbf{t})$ ($r_0 \leq r_1$, $0 \leq d$), and similarly for $\varphi(\mathbf{f}) \leq \varphi(\top) \leq \varphi(\mathbf{t})$. For incomparability of $\bot$ and $\top$: coordinate $i$ gives $t > s$ ($\bot$ higher) while $r$ gives $r_0 < r_1$ ($\top$ higher).

*(2) Information order.* The absolute imaginary magnitudes are $|\mathrm{Im}(\varphi(\bot))| = (t, 0)$, $|\mathrm{Im}(\varphi(\mathbf{f}))| = (|s|, 0)$, $|\mathrm{Im}(\varphi(\mathbf{t}))| = (t, d)$, $|\mathrm{Im}(\varphi(\top))| = (|s|, d)$. Since $|s| > t > 0$ and $d > 0$: $\bot$ has the smallest magnitudes, $\top$ the largest, and $\mathbf{f}$ (large $|i|$, no $|j|$) is incomparable with $\mathbf{t}$ (small $|i|$, large $|j|$).

*(3) Lattice operations.* By construction, $\max_{\mathrm{cw}}(\varphi(\bot), \varphi(\top)) = (r_1, t, d, 0) = \varphi(\mathbf{t})$ and $\min_{\mathrm{cw}}(\varphi(\bot), \varphi(\top)) = (r_0, s, 0, 0) = \varphi(\mathbf{f})$, matching the truth lattice. Similarly, $\max_{|\mathrm{Im}|}(\varphi(\mathbf{f}), \varphi(\mathbf{t})) = (|s|, d) = |\mathrm{Im}(\varphi(\top))|$ and $\min_{|\mathrm{Im}|}(\varphi(\mathbf{f}), \varphi(\mathbf{t})) = (t, 0) = |\mathrm{Im}(\varphi(\bot))|$, matching the information lattice. □

**Remark 11** (Sign-magnitude decoupling)**.** *The embedding exploits the sign degree of freedom in the imaginary components: a* negative *imaginary coordinate is* below *its positive counterpart in component-wise (truth) order, but its* absolute value *is* above*—decoupling truth from information. This is the algebraic mechanism by which the G-lattice accommodates two independent orders on the same space. In Belnap's semantics,* $\mathbf{f}$ *and* $\mathbf{t}$ *carry equal information (*$|\mathrm{Im}|$ *concentrated on different axes) but opposite truth;* $\bot$ *carries minimal information while* $\top$ *carries maximal information, both at neutral truth.*

## 8 The G-Lattice: Context as Group Action

**Definition 12** (Context transformation)**.** *A* context *is a unit quaternion* $q \in \mathrm{SU}(2)$*. The context transformation of a truth value* $a$ *is the conjugation action:*

$$q \cdot a \;=\; q\, a\, \bar{q}.$$

**Definition 13** (G-lattice)**.** *The* quaternionic G-lattice *is the triple* $(L, G, \cdot)$ *where:*

- $L = (V, \mathrm{AND}, \mathrm{OR}, \to, \mathbf{0}, \mathbf{1})$ *is the residuated lattice from Proposition 4, equipped with the involutive negation* NOT *from Remark 5,*
- $G = \mathrm{SU}(2)$ *is the group of unit quaternions,*
- $\cdot$ *is the conjugation action from Definition 12.*

**Proposition 14** (Action properties). *The conjugation action satisfies:*

1. $\mathrm{Re}(q \cdot a) = \mathrm{Re}(a)$ *(truth is objective),*
2. $\|\mathrm{Im}(q \cdot a)\| = \|\mathrm{Im}(a)\|$ *(epistemic weight is conserved),*
3. $\|q \cdot a\| = \|a\|$ *(information is conserved),*
4. $q \cdot (a \cdot_{\mathbb{H}} b) = (q \cdot a) \cdot_{\mathbb{H}} (q \cdot b)$ *(algebra automorphism).*

*Proof.* Properties (1)–(3) follow from equations (1)–(3). Property (4): $q(ab)\bar{q} = (qa\bar{q})(qb\bar{q})$ by associativity and $\bar{q}q = 1$. □

**Proposition 15** (Non-closure of the action on $V$). *The conjugation action is not closed on $V = [0,1] \times [-1,1]^3$. That is, there exist $a \in V$ and $q \in \mathrm{SU}(2)$ such that $q \cdot a \notin V$.*

*Proof.* Let $a = (0.5,\ 1,\ 1,\ 0) \in V$. Then $\|\mathrm{Im}(a)\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$. Conjugation acts on $\mathrm{Im}(a)$ as an $\mathrm{SO}(3)$ rotation preserving $\|\mathrm{Im}\| = \sqrt{2}$. Choose $q \in \mathrm{SU}(2)$ corresponding to the $\mathrm{SO}(3)$ rotation that maps $(1,1,0)$ to $(0,0,\sqrt{2})$. Then $q \cdot a = (0.5,\ 0,\ 0,\ \sqrt{2})$. Since $\sqrt{2} > 1$, the $k$-component exceeds $[-1,1]$, so $q \cdot a \notin V$. □

**Proposition 16** (No domain is simultaneously closed under connectives and the group action). *There is no subset $D \subseteq [0,1] \times \mathbb{R}^3$ that is simultaneously closed under* OR *(component-wise max) and under* $\mathrm{SU}(2)$ *conjugation, unless $D$ is contained in $\{a : \mathrm{Im}(a) = 0\}$.*

*Proof.* Suppose $D$ is closed under both operations and contains some $a$ with $\mathrm{Im}(a) \neq 0$. By $\mathrm{SU}(2)$-closure, $D$ contains all rotations of $\mathrm{Im}(a)$; in particular, it contains elements $a_1, a_2$ with $\mathrm{Re}(a_1) = \mathrm{Re}(a_2)$ and disjoint support in the imaginary components (e.g., $a_1 = (r, \alpha, 0, 0)$ and $a_2 = (r, 0, \alpha, 0)$ for $\alpha = \|\mathrm{Im}(a)\| > 0$). Then $\mathrm{OR}(a_1, a_2) = (r, \alpha, \alpha, 0)$ with $\|\mathrm{Im}\| = \alpha\sqrt{2} > \alpha$. Iterating: OR with a third rotation gives $\|\mathrm{Im}\| = \alpha\sqrt{3}$, and further rotations and ORs produce arbitrarily large $\|\mathrm{Im}\|$. Hence $D$ is unbounded in the imaginary components, contradicting any finite bound. The only escape is $\alpha = 0$, i.e., $\mathrm{Im}(a) = 0$. □

**Remark 17** (Domain separation). *Propositions 15 and 16 establish that the lattice and the group action necessarily live on different domains: the lattice on $V = [0,1] \times [-1,1]^3$, the group action on the ambient space $W = [0,1] \times \mathbb{R}^3 \supseteq V$. This is not a deficiency but a structural fact: the lattice connectives (min, max, Gödel residuum, involutive negation) extend naturally to $W$, and the real-part independence lemma (Lemma 26) holds identically on $W$. Therefore all logical results—tautologies, soundness, completeness, and the commutation theorem—are unaffected by the domain extension. The two structures interact through* Re*, which both preserve.*

**Theorem 18** (Commutation theorem). *Let $f : V^n \to V$ be any connective (AND, OR, NOT, $\to$, or a composition thereof). Then*

$$q \cdot f(a_1, \ldots, a_n) = f(q \cdot a_1, \ldots, q \cdot a_n) \quad \textit{for all } q \in \mathrm{SU}(2)$$

*if and only if $f$ depends only on the real components* $(\mathrm{Re}(a_1), \ldots, \mathrm{Re}(a_n))$.

*Proof.* ($\Leftarrow$) If $f$ depends only on real parts, then $\mathrm{Re}(q \cdot a_m) = \mathrm{Re}(a_m)$ by Proposition 14(1), so $f$ receives the same inputs regardless of $q$. The output has zero imaginary part (since $f$ outputs only a real-part result), and conjugation fixes real quaternions. Hence both sides are equal.

($\Rightarrow$) Suppose $f$ depends on some imaginary component. Choose $a$ such that the imaginary part of $f(a)$ is nonzero. Then there exists $q \in \mathrm{SU}(2)$ that rotates $\mathrm{Im}(f(a))$ to a different direction, while $f(q \cdot a)$ computes min/max on the rotated components of $a$—which generically differs from the rotation of min/max of the original components (since min does not commute with rotation). Hence the two sides differ. □

**Example 19** (Explicit non-commutation). *Let $a = (0.7, 0.3, 0.5, 0.1)$, $b = (0.4, 0.8, 0.2, 0.6)$, and $q = \frac{1}{\sqrt{2}}(1,1,0,0)$. Then $q \cdot \mathrm{AND}(a,b) = (0.4,\ 0.3,\ -0.1,\ 0.2)$ while $\mathrm{AND}(q \cdot a,\ q \cdot b) = (0.4,\ 0.3,\ -0.6,\ 0.2)$; the $j$-components differ ($-0.1$ vs $-0.6$). The real parts agree ($0.4 = 0.4$), confirming that the non-commutation is purely epistemic.*

**Corollary 20** (Boolean logic is context-free)**.** *The sublattice* $\{a \in V : \mathrm{Im}(a) = 0\}$ *is the unique maximal sublattice on which all connectives commute with every context transformation. This is the Boolean/fuzzy restriction.*

**Definition 21** (Context sensitivity)**.** *The* context sensitivity *of a truth value* $a$ *is*

$$\sigma(a) = 2\,\|\,\mathrm{Im}(a)\|.$$

*This equals the maximum displacement* $\max_{q\in \mathrm{SU}(2)} \|q \cdot a - a\|$.

**Remark 22.** *The context sensitivity provides a continuous spectrum:* $\sigma = 0$ *corresponds to Boolean logic (context-free), and increasing* $\sigma$ *gives progressively more context-dependent logics. For truth values in* $V$*, the spectrum is parameterised by* $\|\,\mathrm{Im}(a)\| \in [0, \sqrt{3}]$.

## 9 Soundness

**Definition 23** (Truth predicate)**.** *For a threshold* $\tau \in [0, 1]$*, a truth value* $a$ *is* true *if* $\mathrm{Re}(a) \geq \tau$.

The inference rules of the quaternionic G-lattice are:

**R1**. **Modus ponens:** From $a$ and $a \to b$, infer $b$.
**R2**. **Context shift:** From $a$, infer $q \cdot a$ for any $q \in \mathrm{SU}(2)$.
**R3**. **Context composition:** $q_2 \cdot (q_1 \cdot a) = (q_2 q_1) \cdot a$.

**Theorem 24** (Soundness)**.** *The inference rules* ***R1–R3*** *are sound with respect to the truth predicate of Definition 23.*

*Proof.* **R1** (Modus ponens): If $\mathrm{Re}(a) \geq \tau$ and $\mathrm{Re}(a \to b) \geq \tau$, then either $a_r \leq b_r$ (in which case $b_r \geq a_r \geq \tau$) or $a_r > b_r$ (in which case $\mathrm{Re}(a \to b) = b_r \geq \tau$). In both cases $\mathrm{Re}(b) \geq \tau$. This is the standard soundness argument for the Gödel residuum [Hájek, 1998, Galatos et al., 2007].

**R2** (Context shift): By Proposition 14(1), $\mathrm{Re}(q \cdot a) = \mathrm{Re}(a) \geq \tau$.

**R3** (Context composition): By associativity of quaternion multiplication: $q_2(q_1 a \bar{q}_1)\bar{q}_2 = (q_2 q_1) a (\overline{q_2 q_1})$. □

**Remark 25.** *Soundness decomposes cleanly:* ***R1*** *is inherited from the residuated lattice literature;* ***R2*** *follows from the real-part invariance of conjugation;* ***R3*** *is the group law. No new proof technique is required—the G-lattice soundness is the* product *of two well-understood soundness results.*

## 10 Completeness

**Lemma 26** (Real-part independence)**.** *For every propositional formula* $\varphi$ *built from AND, OR, NOT,* $\to$ *and every valuation* $v : \mathrm{Atoms} \to V$*,*

$$\mathrm{Re}([\![\varphi]\!]_v) \text{ depends only on } (\mathrm{Re}(v(p_1)), \ldots, \mathrm{Re}(v(p_n))).$$

*That is, the imaginary components of the atoms do not affect the real part of the result.*

*Proof.* By structural induction on $\varphi$. For each connective:

$$\begin{aligned}
\mathrm{Re}(\mathrm{AND}(a, b)) &= \min(\mathrm{Re}(a), \mathrm{Re}(b)),\\
\mathrm{Re}(\mathrm{OR}(a, b)) &= \max(\mathrm{Re}(a), \mathrm{Re}(b)),\\
\mathrm{Re}(\mathrm{NOT}(a)) &= 1 - \mathrm{Re}(a),\\
\mathrm{Re}(a \to b) &= \begin{cases} 1 & \text{if } \mathrm{Re}(a) \leq \mathrm{Re}(b),\\ \mathrm{Re}(b) & \text{otherwise.}\end{cases}
\end{aligned}$$

In each case the output's real part is a function of the inputs' real parts alone. The induction step follows by compositionality. □

**Theorem 27** (Completeness)**.** *Let $\varphi$ be a propositional formula built from* $\mathrm{AND}, \mathrm{OR}, \mathrm{NOT}, \to$.

1. *$\varphi$ is a G-lattice tautology ($\mathrm{Re}([\![\varphi]\!]_v) \geq \tau$ for every $v : \mathrm{Atoms} \to V$) if and only if $\varphi$ is a tautology of the $[0,1]$-algebra $(\min, \max, 1-x, \to_G)$, where $\to_G$ is the Gödel residuum.*
2. *For the negation-free fragment ($\varphi$ built from $\mathrm{AND}, \mathrm{OR}, \to$ only), this reduces to Gödel–Dummett completeness [Dummett, 1959, Hájek, 1998].*
3. *For the full fragment (with $\mathrm{NOT}$), the logic is $G^{\sim}$—Gödel logic expanded with an involutive negation—which is complete by Esteva et al. [2000].*

*Proof.* By Lemma 26, $\mathrm{Re}([\![\varphi]\!]_v)$ depends only on the real parts of the atomic valuations. Therefore $\varphi$ is a G-lattice tautology iff it is a tautology of the real-part projection, which is the $[0,1]$-algebra with min, max, the Gödel residuum, and the involutive negation $1-x$.

For the negation-free fragment, this algebra is the standard Gödel chain, and completeness follows from Dummett [1959], Hájek [1998]. For the full fragment, the algebra is a Gödel algebra expanded with involutive negation; completeness of this class ($G^{\sim}$) is established in Esteva et al. [2000]. □

**Remark 28.** *The involutive negation $\mathrm{NOT}(a) = 1 - a$ is* not *the Gödel negation $a \to \mathbf{0}$. In particular, $\mathrm{NOT}(\mathrm{NOT}(a)) = a$ is a tautology (double negation holds), while $(\varphi \to \psi) \to (\mathrm{NOT}(\psi) \to \mathrm{NOT}(\varphi))$ (contraposition) is not. These are characteristic of $G^{\sim}$, not of standard Gödel logic.*

*The imaginary dimensions enrich the logic with context sensitivity, epistemic character, and path-dependent reasoning, but they do not alter which formulas are tautologies. The "new" content of the quaternionic G-lattice is not in its tautologies but in the* structure around *them: the group action, the context spectrum, and the cause–effect chains.*

## 11 Irreducibility of $k$

We first establish a key structural property.

**Proposition 29** (Non-injectivity of projection)**.** *The projection $\pi : H \to [0,1]^3$ defined by $\pi(r,i,j,k) = (r,i,j)$ is not injective on the inhabited region of $H$. That is, there exist states $s_0, s_1 \in H$ with $\pi(s_0) = \pi(s_1)$ and $s_0 \neq s_1$.*

*Proof.* By Axiom 2 (K2), there exists a stagnation state $s_0 = (r_0, i_0, j_0, 0)$ with $r_0, i_0, j_0 > 0$. The liquid-zone state $s_1 = (r_0, i_0, j_0, k_1)$ with any $k_1 > 0$ also lies in $H$ (since $[0,1]^4$ contains all such points). Both states project to $(r_0, i_0, j_0)$ under $\pi$, but $s_0 \neq s_1$ since their $k$-components differ. □

**Theorem 30** (Irreducibility of $k$)**.** *There exists no function $f : [0,1]^3 \to [0,1]$ such that $k = f(r,i,j)$ for all semantic states $(r,i,j,k) \in H$.*

*Proof.* Suppose, for contradiction, that such a function $f$ exists. Let $s_0 = (r_0, i_0, j_0, 0) \in \mathcal{S}$ be a stagnation state (exists by Axiom 2). Then $f(r_0, i_0, j_0) = 0$.

Now consider the state $s_1 = (r_0, i_0, j_0, k_1)$ with $k_1 > 0$ (which exists in $H$ by Proposition 29). Since $f$ determines $k$ from $(r,i,j)$, we must have $f(r_0, i_0, j_0) = k_1 > 0$.

But $f(r_0, i_0, j_0) = 0$ (from $s_0$) and $f(r_0, i_0, j_0) = k_1 > 0$ (from $s_1$) is a contradiction, since $f$ is a function and must assign a unique value to each input. Therefore no such $f$ exists. □

**Corollary 31.** *The four axes $(r, i, j, k)$ are algebraically independent in the sense that no axis can be expressed as a function of the remaining three.*

*Proof.* The irreducibility of $k$ with respect to $(r, i, j)$ is Theorem 30. The irreducibility of each of $r$, $i$, $j$ with respect to the other three follows by an identical argument using the corresponding pathologies: dogma ($i = 0$ with $r, j, k > 0$), delirium ($r = 0$ with $i, j, k > 0$), and pseudoscience

($j = 0$ with $r, i, k > 0$). Each pathology provides a state where one axis is zero while the others are positive, and the liquid zone provides states where all four are positive with the same values on the three non-zero axes. □

**Remark 32** (Strength of the proof)**.** *The proof requires only Axiom K2 (stagnation existence) and the geometric fact that the hypercube* $[0, 1]^4$ *contains both stagnation and liquid-zone states sharing the same* $(r, i, j)$ *projection. No dynamical assumptions, no quaternionic structure, and no threshold mechanism are needed.*

## 12 The Cancellative–Generative Classification

Nine algebraic structures of opposition arise naturally in the prime-algebraic framework. We classify them according to whether they require $k$.

**Definition 33** (Cancellative and generative algebras)**.** *An algebra of opposition is* cancellative *if combining an element with its opposite produces a neutral element (zero, one, or identity matrix). It is* generative *if the combination produces structure that did not exist in either operand.*

**Theorem 34** (Cancellative–generative classification)**.** *The nine algebras of opposition split into two classes:*

***Cancellative*** *(operate within* $(r, i, j)$*, do not require* $k$*):*

1. *Additive inverse:* $a + (-a) = 0$
2. *Multiplicative inverse:* $a \cdot (1/a) = 1$
3. *Cyclic group* $\mathbb{Z}/n\mathbb{Z}$*:* $a + (n - a) \equiv 0$
4. *Rotation* $\mathrm{SO}(2)$*:* $R(\theta) \cdot R(-\theta) = I$

***Generative*** *(require* $k$*):*

5. *Categorical duality:* $(-)^{\mathrm{op}}$ *reverses all arrows*
6. *Kernel/Image:* $\dim(\ker f) + \dim(\operatorname{im} f) = \dim V$
7. *Eigenvalue spectrum:* $Av = \lambda v$ *classifies modes*
8. *Brouwer fixed point:* $\exists\, x^* : f(x^*) = x^*$
9. *Homology/Betti numbers: structure of absence*

*Proof.* **Cancellative algebras (1–4).** Each has the form $a * a^{-1} = e$ where $e$ is an identity element and $*$ is the group operation. This is formally verifiable: given $a$, the result $e$ is uniquely determined by the group axioms—no selection is needed. These algebras operate by producing a neutral element, which is a fixed point of the operation. They model opposition that *annihilates*: wave plus anti-wave yields silence, credit plus debt yields zero. All four are expressible entirely in terms of the content axes $(r, i, j)$: additive inverse operates on $r$ (cancellation of crystallised content), multiplicative inverse on $j$ (normalisation of measurement), cyclic return on $i$ (return of potentiality), and rotation on the $(r, i)$ plane (phase shift without selection).

**Generative algebras (5–9).** The classification criterion for the generative class is the shared structural property: each involves a *meta-operation*—an operation that acts on the structure of a system rather than within it. Theorems 35–40 below prove individually that each algebra's characteristic operation satisfies the three defining properties of $k$: (a) meta-operation on structure, (b) generative, and (c) self-inverting.

*(5) Categorical duality* reverses *all* arrows of a diagram simultaneously—a global structural transformation, not a local computation. It produces "free theorems": any result about products has a dual about coproducts. This requires operating *on* the structure of $(r, i, j)$, not *within* it.

*(6) Kernel/Image* detects what a transformation *destroyed* versus what it *created*. This is a meta-diagnostic: the system auditing its own information loss.

*(7) Eigenvalue spectrum* classifies the *modes* of action ($\lambda > 0$: expansion, $\lambda < 0$: reflection, $\lambda \in \mathbb{C}$: oscillation, $\lambda = 0$: annihilation). Classifying modes requires observing the system's own dynamics from outside.

*(8) Brouwer fixed point* guarantees existence of a point the transformation does not move—the system's own point of stillness. Finding it requires *searching* the space, a selective operation.

*(9) Homology* quantifies the *structure of absence* (how many holes of each dimension). Detecting what is *missing* requires meta-knowledge about the system's own topology.

The common pattern is that each generative algebra requires the system to operate on its own structure, audit its own losses, classify its own modes, search its own space, or detect its own absences—operations characterised by $k$ (recursive selection). A system with $k = 0$ (stagnation) cannot perform any of them, because the content axes $(r, i, j)$ encode *what is known*, not the capacity to interrogate what is known. □

**Theorem 35** (Categorical duality as $k$)**.** *The opposition functor* $(-)^{\mathrm{op}} : \mathbf{Cat} \to \mathbf{Cat}^{\mathrm{op}}$ *is the structural realisation of* $k$ *in the category of categories. Specifically:*

1. *It is a meta-operation on structure (operates on morphisms, not objects)—matching* $k$*'s role as the axis that acts on the other axes.*
2. *It generates without cancelling (no neutral element is produced)—matching the generative character.*
3. $(-)^{\mathrm{op}} \circ (-)^{\mathrm{op}} = \mathrm{Id}$*—matching the involutive character of Axiom 4. See Remark 36 below.*

*Proof.* Properties (1) and (2) follow from the definition of the opposite category: $(-)^{\mathrm{op}}$ reverses all morphisms while preserving objects, and produces genuinely new theorems (the dual of any theorem is a distinct theorem about the opposite structure). Property (3) is standard: $(\mathbf{C}^{\mathrm{op}})^{\mathrm{op}} \cong \mathbf{C}$ for any category $\mathbf{C}$.

The identification with $k$ follows: the only axis that (a) acts on the *structure* of the other axes rather than their content, (b) generates rather than cancels, and (c) is self-inverting, is $k$ by Theorem 34 and Axiom 4. □

**Remark 36** (Sign of self-inversion)**.** *Axiom 4 states* $\mathcal{K} \circ \mathcal{K} = -\mathrm{Id}$ *(matching* $k^2 = -1$ *in quaternion algebra), while the categorical realisation satisfies* $(-)^{\mathrm{op}} \circ (-)^{\mathrm{op}} = \mathrm{Id}$ *(an involution). The essential property shared by both is* self-inversion*: applying selection twice collapses to a determined state. The sign distinguishes two substrates: in* $\mathbb{H}$*, self-inversion produces a* $180°$ *rotation* $(-\mathrm{Id})$*; in* **Cat***, it produces the identity (exact return). Axiom K4 captures the quaternionic version because it preserves the full algebraic structure* $(ij = k,\ jk = i,\ ki = j)$*, which requires* $k^2 = -1$*. The categorical realisation satisfies the weaker involutive form—it captures the generative and self-inverting character of* $k$ *but not the rotational sign.*

**Theorem 37** (Kernel/image as $k$)**.** *The rank-nullity decomposition* $V = \ker(f) \oplus \ker(f)^{\perp}$ *of a linear map* $f : V \to W$ *(on finite-dimensional inner product spaces) is the structural realisation of* $k$ *in linear algebra:*

1. *It is a meta-operation: testing* $v \in \ker(f)$ *requires evaluating* $f(v) = 0$*, an examination of the map's own behaviour.*
2. *It generates without cancelling: the dimensions* $\dim \ker f$ *and* $\operatorname{rank} f$ *are global invariants invisible to any single evaluation* $f(v)$*.*
3. *The complementary-projection map* $P \mapsto \mathrm{Id} - P$ *is an involution:* $\mathrm{Id} - (\mathrm{Id} - P) = P$*.*

*Proof.* (1) Membership $v \in \ker(f)$ requires comparing $f(v)$ against a criterion $(= 0)$: a diagnostic on $f$'s output, not a computation within $V$. (2) The rank-nullity identity $\dim \ker f + \operatorname{rank} f = \dim V$ cannot be determined from fewer than $\dim V$ linearly independent evaluations—assembling such a spanning set is itself a selective ($k$) operation. (3) Immediate: $C(P) = \mathrm{Id} - P$ satisfies $C \circ C = \mathrm{Id}$.

A system with $k = 0$ can compute $f(v)$ for given $v$ but cannot test whether $f(v) = 0$ (no meta-diagnostic) and therefore cannot partition $V$ into kernel and image. □

**Theorem 38** (Eigenvalue spectrum as $k$)**.** *For a normal operator* $A$ *on a finite-dimensional inner product space, the spectral decomposition* $A = \sum_i \lambda_i P_i$ *is the structural realisation of* $k$ *in operator algebra:*

1. *It is a meta-operation: the characteristic polynomial* $\det(A-\lambda I)=0$ *depends on all entries of* $A$ *simultaneously.*
2. *It generates without cancelling: the eigenvalues classify* $A$*'s modes (expansion, contraction, oscillation, annihilation).*
3. *The adjoint map* $A \mapsto A^*$ *is an involution on operators* ($A^{**}=A$) *inducing conjugation* $\lambda \mapsto \bar{\lambda}$ *on spectra.*

*Proof.* (1) The characteristic polynomial $p_A(\lambda)=\det(A-\lambda I)$ is a global function of $A$'s matrix entries; no single evaluation $Av$ determines any eigenvalue. (2) The spectrum $\sigma(A)$ classifies qualitative dynamics on each eigenspace: a system limited to content axes sees $Av$ as a vector, not as $\lambda v$ for a mode $\lambda$. (3) For normal $A=U\Lambda U^*$, $A^*=U\bar{\Lambda}U^*$, giving $\sigma(A^*)=\overline{\sigma(A)}$, and $(A^*)^*=A$.

A system with $k=0$ can compute $Av$ but cannot extract the eigenvalues—it sees individual outputs without classifying the modes that generate them. □

**Theorem 39** (Brouwer fixed point as $k$)**.** *The fixed-point operator* $\mathrm{Fix}(f)=\{x\in D : f(x)=x\}$ *for continuous* $f: D\to D$ *on compact convex* $D\subset\mathbb{R}^n$ *is the structural realisation of* $k$ *in continuous dynamics:*

1. *It is a meta-operation:* $f(x)=x$ *compares the system's output with its input—a self-referential diagnostic.*
2. *It generates without cancelling: Brouwer's theorem guarantees* $\mathrm{Fix}(f)\neq\emptyset$*, a topological existence result not determined by finitely many evaluations of* $f$*.*
3. *The reflection* $f\mapsto 2\mathrm{Id}-f$ *is a fixed-point-preserving involution:* $\mathrm{Fix}(f)=\mathrm{Fix}(2\mathrm{Id}-f)$*, and* $2\mathrm{Id}-(2\mathrm{Id}-f)=f$*.*

*Proof.* (1) The test $f(x)\stackrel{?}{=}x$ requires simultaneous access to $f$ and Id—an operation on $f$'s structure, not within its domain. (2) Brouwer's theorem follows from the Lefschetz fixed point theorem ($L(f)\neq 0$ for contractible $D$): a global topological invariant, not a local computation. (3) If $f(x^*)=x^*$ then $(2x^*-f(x^*))=x^*$, so $x^*\in\mathrm{Fix}(2\mathrm{Id}-f)$, and conversely. The composition $2\mathrm{Id}-(2\mathrm{Id}-f)=f$ is immediate.

A system with $k=0$ can evaluate $f(x)$ pointwise but cannot compare $f(x)$ with $x$ (no self-referential test) and therefore cannot locate fixed points. □

**Theorem 40** (Homology as $k$)**.** *For a simplicial complex* $K$*, the Betti numbers* $\beta_n=\dim H_n(K)=\dim(\ker\partial_n/\operatorname{im}\partial_{n+1})$ *are the structural realisation of* $k$ *in algebraic topology:*

1. *It is a meta-operation: computing* $H_n$ *requires comparing* $\ker\partial_n$ *against* $\operatorname{im}\partial_{n+1}$*—an analysis of the boundary operators' own structure.*
2. *It generates without cancelling: the Betti numbers quantify the "structure of absence" (independent* $n$*-cycles that do not bound).*
3. *Poincaré duality for closed oriented* $n$*-manifolds,* $H_k(M)\cong H_{n-k}(M)$*, is a dimension-swapping involution:* $k\mapsto n-k\mapsto k$*.*

*Proof.* (1) The quotient $\ker\partial_n/\operatorname{im}\partial_{n+1}$ asks "which cycles are boundaries?"—a meta-question about the relationship between consecutive chain groups, not a computation within any single group. (2) Betti numbers detect what is *absent*: $\beta_1$ counts independent loops, $\beta_2$ counts independent voids. This is information about holes in $K$, not about simplices present in $K$. (3) For closed oriented $n$-manifolds, $\beta_k=\beta_{n-k}$, and the dimension map $k\mapsto n-k$ satisfies $n-(n-k)=k$.

A system with $k=0$ can compute boundaries $\partial\sigma$ of individual simplices but cannot determine which cycles bound—it sees local boundary relations without the global analysis distinguishing homologous from trivial cycles. □

**Remark 41** (Involution forms)**.** *Each generative algebra realises* $k$*'s self-inverting character through a different involution: complementary projection* ($P\mapsto\mathrm{Id}-P$) *for kernel/image, operator adjunction* ($A\mapsto A^*$) *for eigenvalues, dynamical reflection* ($f\mapsto 2\mathrm{Id}-f$) *for fixed points,*

*and Poincaré duality ($H_k \leftrightarrow H_{n-k}$) for homology. Like Remark 36, these are substrate-specific realisations of the same structural property: double application returns to the original.*

## 13 Pathologies as Axis Collapses

Each kingdom's absence produces a diagnosable pathology:

**Dogma** ($i = 0$)

$$\text{Dogma} = (r > 0,\ i = 0,\ j > 0,\ k \geq 0) \tag{17}$$

Crystallised, verified, possibly even selective—but without potentiality. No "what if?" is permitted. Truth frozen. Ice that has forgotten it was once water.

**Delirium** ($r = 0$)

$$\text{Delirium} = (r = 0,\ i > 0,\ j \geq 0,\ k \geq 0) \tag{18}$$

Pure potentiality without crystallisation. Heat without container. The artist with a thousand visions who finishes nothing.

**Pseudoscience** ($j = 0$)

$$\text{Pseudoscience} = (r > 0,\ i > 0,\ j = 0,\ k \geq 0) \tag{19}$$

Structure and imagination without verification. "It sounds right" and "everything fits" but nobody has measured anything. The paper that diagnostics detects: elegant framework, zero empirical contact.

**Stagnation** ($k = 0$)

$$\text{Stagnation} = (r > 0,\ i > 0,\ j > 0,\ k = 0) \tag{20}$$

Crystallised, imaginative, verified—but never pruning, never rotating, never advancing. The committee that discusses endlessly without deciding. Evolution without natural selection: variation without fitness.

**Definition 42** (Liquid Zone)**.** *The **liquid zone** is the region of the semantic hypercube where all four axes are simultaneously non-zero:*

$$\mathcal{Z} = \{s \in [0,1]^4 : r > 0 \ \wedge\ i > 0 \ \wedge\ j > 0 \ \wedge\ k > 0\} \tag{21}$$

*Phase transitions are trajectories that enter or exit $\mathcal{Z}$ by collapsing or activating an axis.*

### 13.1 Topological Properties

**Proposition 43** (Openness)**.** *$\mathcal{Z}$ is open in the relative topology of $H = [0,1]^4$.*

*Proof.* For each coordinate $x \in \{r, i, j, k\}$, the set $\{s \in H : x > 0\}$ is open in $H$ (it is $H \cap \{x > 0\}$, and $\{x > 0\}$ is open in $\mathbb{R}^4$). Since $\mathcal{Z}$ is the intersection of four such open sets, it is open. □

**Proposition 44** (Convexity and path-connectedness)**.** *$\mathcal{Z}$ is convex and hence path-connected.*

*Proof.* Let $s_1, s_2 \in \mathcal{Z}$ and $t \in [0,1]$. Then $s_t = (1-t)s_1 + ts_2$ satisfies: each coordinate of $s_t$ is a convex combination of two strictly positive numbers in $[0,1]$, hence strictly positive and in $[0,1]$. Therefore $s_t \in \mathcal{Z}$, and $\mathcal{Z}$ is convex. Convexity implies path-connectedness. □

**Proposition 45** (Disjointness of pathologies)**.** *The four pathological regions are pairwise disjoint.*

*Proof.* Each pathology sets a *different* axis to zero while requiring at least two others to be strictly positive. For any pair of pathologies, at least one requires a coordinate to be positive that the other requires to be zero (or vice versa), so their intersection is empty. Explicitly: $\mathcal{D}_{\text{dogma}} \cap \mathcal{D}_{\text{delirium}} = \emptyset$ because dogma requires $r > 0$ while delirium requires $r = 0$. $\mathcal{D}_{\text{dogma}} \cap \mathcal{D}_{\text{pseudo}} = \emptyset$ because dogma requires $j > 0$ while pseudoscience requires $j = 0$. $\mathcal{D}_{\text{dogma}} \cap \mathcal{D}_{\text{stag}} = \emptyset$ because dogma requires $i = 0$ while stagnation requires $i > 0$. $\mathcal{D}_{\text{delirium}} \cap \mathcal{D}_{\text{pseudo}} = \emptyset$ because delirium requires $r = 0$ while pseudoscience requires $r > 0$. $\mathcal{D}_{\text{delirium}} \cap \mathcal{D}_{\text{stag}} = \emptyset$ because delirium requires $r = 0$ while stagnation requires $r > 0$. $\mathcal{D}_{\text{pseudo}} \cap \mathcal{D}_{\text{stag}} = \emptyset$ because pseudoscience requires $j = 0$ while stagnation requires $j > 0$. □

**Proposition 46** (Complementarity)**.** *$\mathcal{Z} \cap \mathcal{D}_\alpha = \emptyset$ for each pathology $\alpha$. The liquid zone is disjoint from all pathological regions.*

*Proof.* Every state in $\mathcal{Z}$ has all four coordinates strictly positive. Every pathological state has at least one coordinate equal to zero. □

## 14 The Quaternionic Product on $S^3$

The four axes $\{1, i, j, k\}$ mirror the basis of Hamilton's quaternions $\mathbb{H}$, with the multiplication rules:

$$i^2 = j^2 = k^2 = ijk = -1 \tag{22}$$

$$ij = k, \quad jk = i, \quad ki = j \tag{23}$$

Each rule receives a semantic interpretation:

1. $i^2 = -1$: Potentiality interacting with itself produces inversion. A dream that dreams itself collapses or inverts.
2. $j^2 = -1$: Verification verifying itself without content produces negation. A measurement of a measurement, absent anything to measure, is vacuous.
3. $k^2 = -1$: Selection selecting itself produces inversion. Recursion without base case is infinite regress. (Axiom 4.)
4. $ij = k$: Potentiality plus directionality produces selection. Imagining *where to go* generates the capacity to choose. (Axiom 3.)
5. $jk = i$: Directionality plus selection produces potentiality. Constraint enables creativity.
6. $ki = j$: Selection plus potentiality produces directionality. Choosing among possibilities creates order.

**Remark 47.** *The non-commutativity of quaternion multiplication ($ij \neq ji$) has a direct semantic reading: imagining first and then measuring produces a different outcome than measuring first and then imagining. The order of ontological operations matters.*

**Theorem 48** (Semantic quaternionic product)**.** *For semantic states $s_1, s_2 \in \mathbb{R}^4 \setminus \{0\}$, define $\hat{s} = s/\|s\|$ and $s_1 \otimes s_2 = \hat{s}_1 \cdot_{\mathbb{H}} \hat{s}_2$, where $\cdot_{\mathbb{H}}$ is Hamilton multiplication. This product:*

1. *is closed on $S^3$,*
2. *is associative,*
3. *admits inverses,*
4. *satisfies the basis relations: $ij = k$, $jk = i$, $ki = j$, $i^2 = j^2 = k^2 = ijk = -1$,*
5. *is non-commutative ($ij \neq ji$), and*

*6. respects layer recovery: products of real-subspace elements remain in the real subspace.*

*Proof.* Write $\hat{s}_j = s_j/\|s_j\|$ for the unit normalisation.

*(1) Closure.* $\|\hat{s}_1 \cdot_{\mathbb{H}} \hat{s}_2\| = \|\hat{s}_1\|\,\|\hat{s}_2\| = 1 \cdot 1 = 1$, by the multiplicativity of the quaternion norm.

*(2) Associativity.* $\hat{s}_1 \otimes (\hat{s}_2 \otimes \hat{s}_3) = \hat{s}_1 \cdot_{\mathbb{H}} (\hat{s}_2 \cdot_{\mathbb{H}} \hat{s}_3) = (\hat{s}_1 \cdot_{\mathbb{H}} \hat{s}_2) \cdot_{\mathbb{H}} \hat{s}_3 = (\hat{s}_1 \otimes \hat{s}_2) \otimes \hat{s}_3$, since Hamilton multiplication is associative and the normalisation of a unit quaternion is the identity.

*(3) Inverses.* For any unit quaternion $\hat{s}$, $\hat{s}^{-1} = \overline{\hat{s}}$ (the quaternion conjugate), and $\hat{s}\,\overline{\hat{s}} = \|\hat{s}\|^2 = 1$.

*(4) Basis relations.* These are the defining relations of the quaternion algebra $\mathbb{H} = \langle 1, i, j, k \mid i^2 = j^2 = k^2 = ijk = -1 \rangle$.

*(5) Non-commutativity.* $ij = k$ while $ji = -k$, so $ij \neq ji$.

*(6) Real subspace.* If $\mathrm{Im}(\hat{s}_1) = \mathrm{Im}(\hat{s}_2) = 0$, then $\hat{s}_1, \hat{s}_2 \in \{\pm 1\} \subset \mathbb{R} \subset \mathbb{H}$, and their product is $\pm 1$, which is real.

All six properties are additionally verified computationally at machine precision ($\varepsilon < 10^{-15}$) over 10,000 random pairs and all 72 semantic primitives from Ornelas Brand [2026b]. □

## 15 Relation to Information Theory

Shannon [1948] defined information along a single axis: certainty/uncertainty. A classical bit is $\{0, 1\}$ and entropy $H = -\sum p_i \log p_i$ measures ignorance. But this captures only *resolved* information—the projection onto $r$.

**Proposition 49** (Shannon as special case)**.** *Shannon information theory is the restriction of $\mathcal{O}$ to:*

$$\textit{Shannon} = \mathcal{O}\big|_{i=0,\ j=1,\ k=0} = \textit{projection onto } r \tag{24}$$

*The conditions are: no potentiality (all information is resolved), full verifiability (all information is measurable), and no recursive selection (the channel is passive).*

*Proof.* Under the Shannon restriction: $i = 0$ (all information is resolved), $j = 1$ (all information is fully verifiable), $k = 0$ (the channel is passive—it transmits but does not select). Shannon's entropy operates on probability distributions over resolved, measurable states with no selective agency. The condition $k = 0$ explains why Shannon's theory has no concept of a channel that *chooses* what to transmit: the selective axis is collapsed. □

The natural extension follows the algebraic progression of number systems:

| Number system | Operator | Information captured |
|---|---|---|
| $\mathbb{R}$ (reals) | $\{0, 1\}$ | Classical bit: presence/absence |
| $\mathbb{C}$ (complex) | $(r, i)$ | Information with potentiality |
| $\mathbb{H}$ (quaternions) | $(r, i, j, k)$ | Information with potentiality, verifiability, and selection |

## 16 Epistemic Dimensions Are Not Decoration

A potential objection is that since $\mathrm{Re}(a)$ is always context-invariant, the imaginary dimensions $(i, j, k)$ are "decoration" that do not affect logical reasoning. We present three arguments against this objection.

### 16.1 Operational distinguishability

**Definition 50** (Decision function)**.** *A* decision function *is any map $D : V \to \mathcal{A}$ from truth values to an action space $\mathcal{A}$ that depends on the full quaternionic truth value, not only on* $\mathrm{Re}(a)$*.*

**Proposition 51.** *If $D$ depends on* $\mathrm{Im}(a)$*, then there exist truth values $a, b$ with* $\mathrm{Re}(a) = \mathrm{Re}(b)$ *such that $D(a) \neq D(b)$.*

*Proof.* Take $a = (r, \alpha, 0, 0)$ and $b = (r, 0, \alpha, 0)$ for any $r \in (0, 1)$ and $\alpha \neq 0$. Then $\mathrm{Re}(a) = \mathrm{Re}(b) = r$ but $\mathrm{Im}(a) \neq \mathrm{Im}(b)$, and by assumption $D$ distinguishes them. □

**Example 52** (The anchor argument)**.** *A goal is scored: $a = (0.9, 0.6, 0.7, 0.5)$. Both teams agree on the fact (*$\mathrm{Re} = 0.9$*). Let $q_{\mathrm{red}}$ be a context rotation (a different "anchor"):*

$$q_{\mathrm{red}} \cdot a = (0.9,\ 0.6,\ -0.7,\ -0.5).$$

*The truth is identical. The epistemic character differs: one team reads positive verification ($+j$) and celebrates; the other reads negative verification ($-j$) and laments. The decision (celebrate or lament) depends on $(i, j, k)$, not only on $r$.*

### 16.2 The panorama

Averaging over all contexts yields a pure-real result:

$$\int_{\mathrm{SU}(2)} q \cdot a \, d\mu(q) = \big(\mathrm{Re}(a),\ 0,\ 0,\ 0\big),$$

where $\mu$ is the Haar measure on SU(2). This is the "view from nowhere"—Boolean logic. Agents, however, do not live in the panorama; they are situated in specific contexts. The imaginary dimensions encode the *situated perspective* that determines behaviour.

### 16.3 Epistemic quality

Two propositions with the same truth ($\mathrm{Re} = 0.7$) but different epistemic profiles carry different information:

- (0.7, 0.8, 0.1, 0.2): high potentiality, low verification—a *hypothesis*.
- (0.7, 0.1, 0.8, 0.2): low potentiality, high verification—an *observation*.

A rational agent treats hypotheses and observations differently, even at equal truth values. The dimensions encode *epistemic quality*, which is decision-relevant.

## 17 Cause–Effect Chains

Since SU(2) is non-abelian, the composition of context changes is order-dependent:

$$q_2 \cdot (q_1 \cdot a) \neq q_1 \cdot (q_2 \cdot a) \quad \text{in general.}$$

All orderings preserve the same $\mathrm{Re}(a)$, but produce different epistemic signatures Im. The process retains *memory of the path taken*: the order of perspective changes leaves a distinct epistemic trace.

**Proposition 53** (Path dependence)**.** *For $n$ distinct context changes $q_1, \ldots, q_n \in \mathrm{SU}(2)$, the $n!$ orderings of application to a truth value $a$ all yield the same* $\mathrm{Re}(a)$ *but generically yield $n!$ distinct imaginary parts.*

*Proof.* Re is invariant under every conjugation (iterated application of Proposition 14(1)). For the imaginary part, distinct orderings correspond to distinct elements of SU(2) (since the group is non-abelian), which generically produce distinct rotations of $\mathrm{Im}(a)$. □

This formalises the notion that "imagining then verifying" and "verifying then imagining" leave different epistemic signatures, even when they agree on the truth. The non-commutativity is the algebraic encoding of irreversible cause–effect chains.

## 18 Embedding the Prime Algebra into the G-Lattice

The prime-algebraic framework of Ornelas Brand [2026c] encodes each concept $C$ as a product of primes $\Phi(C) = \prod_{i=1}^{72} p_i^{s_i}$, $s_i \in \{0,1\}$, where the 72 primitives are distributed across six algebraic layers. The binary exponent vector $\mathbf{s}(C) = (s_1, \ldots, s_{72}) \in \{0,1\}^{72}$ turns divisibility into componentwise $\leq$, gcd into componentwise min, and lcm into componentwise max [Birkhoff, 1967].

**Definition 54** (Layer projection)**.** *Partition the 72 indices into layer sets $L_1, \ldots, L_6$ ($|L_1| = 3$, $|L_2| = 8$, $|L_3| = 13$, $|L_4| = 21$, $|L_5| = 23$, $|L_6| = 4$). Define axis-contributing sets by the semantic content of Table 1:*

$$\begin{aligned} R &= L_1 \cup L_2 \cup L_4 \cup L_6 && (|R| = 36), \\ I &= L_4 \cup L_6 && (|I| = 25), \\ J &= L_3 \cup L_5 \cup L_6 && (|J| = 40), \\ K &= L_6 && (|K| = 4). \end{aligned}$$

*The* layer embedding *is $e\colon \{0,1\}^{72} \to [0,1]^4$,*

$$e(\mathbf{s}) \;=\; \Big( \frac{\sum_{i\in R} s_i}{|R|}, \frac{\sum_{i\in I} s_i}{|I|}, \frac{\sum_{i\in J} s_i}{|J|}, \frac{\sum_{i\in K} s_i}{|K|} \Big).$$

*For the G-lattice domain $V = [0,1] \times [-1,1]^3$, rescale the imaginary components: $\hat{e} = (e_r,\, 2e_i - 1,\, 2e_j - 1,\, 2e_k - 1)$.*

**Theorem 55** (Synthesis embedding)**.** *Let $A, B$ be concepts with exponent vectors $\mathbf{s}(A), \mathbf{s}(B)$.*

1. ***Monotonicity.*** *$\Phi(A) \mid \Phi(B)$ implies $e(\mathbf{s}(A)) \leq e(\mathbf{s}(B))$ componentwise.*
2. ***Lattice compatibility.*** *$e(\mathbf{s}(\mathrm{lcm})) \geq \max\big(e(\mathbf{s}(A)), e(\mathbf{s}(B))\big)$ componentwise, where* lcm *denotes $\mathrm{lcm}(\Phi(A), \Phi(B))$.*
3. ***Strict synthesis.*** *Define the* interiors *$\mathrm{Int}(A \mid B) = \{i : s_i(A) = 1,\, s_i(B) = 0\}$ and $\mathrm{Int}(B \mid A) = \{i : s_i(B) = 1,\, s_i(A) = 0\}$. The inequality in (2) is strict on axis $X$ if and only if both interiors have $X$-support:*

$$e(\mathbf{s}(\mathrm{lcm}))_X > \max(e(\mathbf{s}(A))_X,\, e(\mathbf{s}(B))_X) \iff \mathrm{Int}(A \,|\, B) \cap X \neq \emptyset \text{ and } \mathrm{Int}(B \,|\, A) \cap X \neq \emptyset.$$

*Proof.* **(1)** $\Phi(A) \mid \Phi(B)$ means $s_i(A) \leq s_i(B)$ for all $i$, so $\sum_{i\in X} s_i(A) \leq \sum_{i\in X} s_i(B)$ for every axis set $X$.

**(2)** $s_i(\mathrm{lcm}) = \max(s_i(A), s_i(B)) \geq s_i(A)$ for all $i$, so $\sum_{i\in X} \max(s_i(A), s_i(B)) \geq \sum_{i\in X} s_i(A)$, and similarly for $B$. Dividing by $|X|$ gives $e(\mathbf{s}(\mathrm{lcm}))_X \geq \max(e_X(A), e_X(B))$.

**(3)** Let $A_X = \{i \in X : s_i(A) = 1\}$ and $B_X = \{i \in X : s_i(B) = 1\}$. Then $e(\mathbf{s}(\mathrm{lcm}))_X = |A_X \cup B_X|/|X|$ and $\max(e_X(A), e_X(B)) = \max(|A_X|, |B_X|)/|X|$. The union strictly exceeds the larger set iff neither is a subset of the other: $A_X \not\subseteq B_X$ and $B_X \not\subseteq A_X$. The first holds iff $\mathrm{Int}(A \,|\, B) \cap X \neq \emptyset$; the second iff $\mathrm{Int}(B \,|\, A) \cap X \neq \emptyset$. □

**Remark 56** (Degenerate pairs)**.** *When $\Phi(A) \mid \Phi(B)$, the interior $\mathrm{Int}(A \mid B)$ is empty and synthesis reduces to $\mathrm{lcm} = \Phi(B)$: no axis shows strict increase because $A$ is already contained in $B$. Three of the fourteen duality pairs from Ornelas Brand [2026c] exhibit this degeneracy (their dependency chains are in a subset relation); the remaining eleven all show strict increase on every axis with bilateral interior support, as predicted by Theorem 55(3).*

## 19 Computational Verification

All results were verified computationally using NumPy in Python. Random test cases were drawn uniformly from the appropriate domains.

Table 2: Summary of computational verification.

| **Test** | **N** | **Violations** | **Max error** |
|---|---|---|---|
| Specialisations (4 exact + 2 interpretive) | — | 0 | exact |
| De Morgan laws (extended domain) | 50 000 | 0 | 0 |
| Re invariance under conjugation | 50 000 | 0 | $5.55 \times 10^{-16}$ |
| $\lVert \mathrm{Im} \rVert$ invariance under conjugation | 10 000 | 0 | $6.66 \times 10^{-16}$ |
| Algebra automorphism | 10 000 | 0 | $< 10^{-12}$ |
| Commutation (Boolean AND) | 10 000 | 0 | $3.33 \times 10^{-16}$ |
| Non-commutation (full AND) | 10 000 | 9 634 | 1.16 |
| Non-commutation (full OR) | 10 000 | 9 637 | 1.23 |
| Non-commutation (NOT) | 10 000 | 10 000 | 2.73 |
| Modus ponens (all contexts) | 50 000 | 0 | exact |
| Context shift preserves truth | 50 000 | 0 | exact |
| Re of compound formula invariant | 50 000 | 0 | $9.99 \times 10^{-16}$ |
| Domain closure ($r$ in $[0, 1]$) | 50 000 | 0 | exact |
| Decision distinguishability | 50 000 | — | 49.6% changed |
| Panorama ($\mathrm{Im} \to 0$) | 10 000 | — | $\lVert \mathrm{Im} \rVert < 0.005$ |
| Re-independence (all connectives) | 100 000 | 0 | 0 |
| Re-independence (7 compounds) | 100 000 | 0 | 0 |
| $G^{\sim}$ tautologies (6 formulas) | 100 000 | 0 | confirmed |
| Non-tautologies (4 formulas) | 100 000 | 0 | confirmed |
| *Embedding (Section 18)* | | | |
| Layer consistency (P1) | 4 | 0 | exact |
| Divisibility $\Rightarrow$ order (P3) | 10 000 | 0 | exact |
| lcm $\geq$ max (P4) | 10 000 | 0 | exact |
| Bilateral synthesis (14 pairs) | 14 | 0 | exact |

Source code is available at `scripts/{connectives, context_bridge, verify_resolutions, verify_completeness}.py` in the companion repository.

## 20 Related Work

Table 3: Comparison with existing algebraic structures for logic.

| Structure | Source | Relation |
|---|---|---|
| Residuated lattice | Galatos et al. [2007] | Layer 1 of our structure |
| BL-algebra | Hájek [1998] | Our lattice under fuzzy restriction |
| MV-algebra | Cignoli et al. [2000] | Related to Łukasiewicz specialisation |
| Quaternionic fuzzy sets | Dai [2023] | Quaternion-valued membership; no group action, no logic |
| Quantale | Mulvey [1986] | Complete lattice + monoid; ours uses non-abelian group |
| Quantum logic | Birkhoff & von Neumann [1936] | Projections on Hilbert spaces, not t-norms |
| Substructural logic | Restall [2000] | Non-commutative connectives, no group action |

The closest antecedent is Dai [2023], who defines quaternionic fuzzy *sets* with quaternion-valued membership functions bounded by unit modulus. The present work differs in three respects: (i) we define a propositional *logic* with inference rules, soundness, and completeness, not only set operations; (ii) we add a group action that separates truth from epistemic character via the commutation theorem; (iii) our domain is component-wise bounded ($[0,1] \times [-1,1]^3$) rather than modulus-bounded.

More broadly, the distinguishing feature of the present work is that the group action is not applied to the connectives (which remain commutative) but to the *truth values themselves*, with a clean invariance theorem separating objective truth from perspectival epistemic character.

## 21 Conclusion

We have constructed a quaternionic operator $\mathcal{O} = \{0, 1, +, i, j, k\}$ from which six algebraic layers are recoverable as dimensional projections—four exactly (Boolean, Gödel fuzzy, Gödel–Kripke modal, and Kleene trivalent) and two interpretively (ordinal and probabilistic)—and shown that it supports a G-lattice—a residuated lattice equipped with an SU(2) group action—that genuinely unifies Boolean and fuzzy frameworks.

The unification is not mere containment: it is mediated by a commutation theorem (Theorem 18) that characterises Boolean logic as the context-free special case. The fourth axis $k$ is proven irreducible (Theorem 30), and its presence divides the nine known algebras of opposition into cancellative and generative classes (Theorem 34), with formal $k$-embeddings for all five generative algebras (Theorems 35–40). Each axis's absence produces a diagnosable pathology, and the region where all four axes are simultaneously active—the liquid zone—is open, convex, and path-connected.

Completeness reduces to $G^{\sim}$ (Gödel logic with involutive negation) via the real-part independence lemma. The imaginary dimensions do not alter tautologies—they enrich the surrounding structure with context sensitivity, path-dependent epistemic chains, and operationally distinguishable epistemic dimensions. A layer-based embedding (Theorem 55) connects the prime-algebraic framework of Ornelas Brand [2026c] to the G-lattice coordinates, with a bilateral synthesis theorem characterising exactly when duality synthesis produces strict increase on each axis.

The framework developed here is extended in three companion papers: Ornelas Brand [2026d] investigates whether Hamilton multiplication yields a genuinely non-commutative conjunction,

proving a trilemma that rules out such constructions on the semantic domain; Ornelas Brand [2026e] develops the synthesis operation on the prime algebra and shows that opposites generate new truth at higher algebraic layers through an irreducibly second-order gap operation; and Ornelas Brand [2026f] formalises pre-logical states—points in $V$ where the truth axis is undefined—and proves that information emerges as the compound concept of the pre-logical/logical duality, with substrate independence demonstrated across four domains.

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